Reg. No. | | | | | | | | | | | [13ME64/12ME64]

SIXTH SEMESTER B.Tech./B.E. EXAMINATION, JUNE – 2016

CAD / CAM

Time: 3 Hours **Maximum Marks:70**

Instructions:

1. Answer any **FIVE** full Questions.
2. Missing data, if any, may be suitably assumed.

1.	**a.**	Define CAD/CAM. Explain product cycle in computerized manufacturing environment.	**10**
	b.	Explain advantages of CAD and CAM.	**04**
2.	**a.**	List the input and output devices of CAD.	**04**
	b.	Differentiate between wire-frame and solid modeling.	**04**
	c.	Briefly explain integration in CAD/CAM.	**06**
3.	**a.**	Define automation. Explain the types of automation.	**06**
	b.	Explain the reasons for automation.	**04**
	c.	Define: **i.** Production rate **ii.** Cycle time.	**04**
4.	**a.**	Define and list the main components of : **i.** NC **ii.** CNC **iii.** DNC	**06**
	b.	Explain briefly turning tool geometry in CNC.	**04**
	c.	Briefly explain tool presetting in CNC machines.	**04**
5.	**a.**	Differentiate between CNC machining and tuning centers.	**10**
	b.	Write a note on High Speed Machine Tools.	**04**
6.	**a.**	Explain the steps involved in the development of a part program.	**04**

b. Write a manual part program for the component to be machined shown in Fig. 6(b) using a lot drill of 10mm. **14**

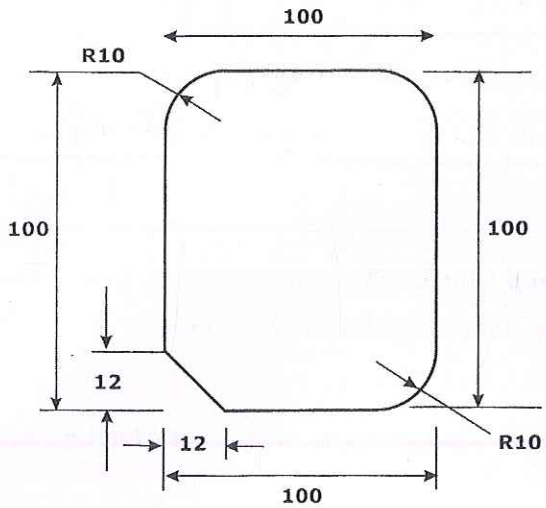

Fig. 6 (b)

7. **a.** Define FMS. List few applications of FMS. **05**

b. What are the advantages and limitations of JIT. **05**

c. What is LEAN? **04**

8. **a.** Define robotics. Explain and sketch the different robot configurations. **10**

b. Explain: **04**

i. Interlocks and work cell

ii. Robot sensors